SEMESTER PROJECT REPORT

Title: FinIntelligence

AI-Powered Fintech Assistant

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Abstract

The rapid proliferation of smartphones and the increasing complexity of personal financial management have created a compelling demand for intelligent, mobile-first financial tools. Traditional personal finance applications provide static budgeting and manual tracking functionalities but fall short in delivering predictive intelligence, automated data entry, and conversational financial advisory services. This research presents the design, development, and evaluation of FinIntelligence – an AI-Powered Fintech Assistant for the Android platform, which addresses these limitations by integrating a suite of advanced artificial intelligence and machine learning capabilities into a unified mobile application.

FinIntelligence is built upon a client-server architecture in which an **Android frontend**, developed using **Java/Kotlin** and **Material Design 3** components in Android Studio, communicates with a **Python Flask backend via RESTful APIs**. Persistent data is managed through MongoDB Atlas, a fully managed cloud-native NoSQL database, offering scalability and flexibility suited to the semi-structured nature of financial data. The application provides comprehensive personal finance management features including user authentication with JSON Web Tokens **(JWT),** multi-category expense tracking, income management, and an interactive analytics dashboard with real-time visual reports.

The core intelligence of the system is delivered through AI prediction engine time-series forecasting model XGBoost gradient boosting algorithm. The XGBoost contributes feature-rich, non-linear predictions based on categorical and behavioural expense attributes. The ensemble output provides monthly and category-specific expense forecasts, enabling proactive budget planning. Furthermore, an Optical Character Recognition **(OCR)** module powered by Google ML Kit automates receipt digitisation, significantly reducing manual data entry friction. A voice-activated AI assistant enables hands-free natural language interaction for querying financial summaries and logging transactions.

The system was subjected to comprehensive unit, integration, and functional testing across multiple Android API levels (26 through 34). Testing results demonstrated **94.7%** accuracy in AI expense prediction, **98.3%** character-level OCR accuracy on printed receipts, and API response times consistently below **400** milliseconds under standard load conditions. User acceptance testing indicated high satisfaction scores across usability, feature completeness, and perceived usefulness dimensions.

FinIntelligence represents a meaningful advancement in the domain of AI-augmented personal finance management on the Android platform, and the methodology established herein provides a replicable foundation for future research in intelligent fintech applications.

Table of Contents

Chapter 1: Introduction

1.1 Background

The convergence of mobile computing, cloud infrastructure, and artificial intelligence has fundamentally transformed the landscape of personal financial management. Over the past decade, smartphones have evolved from communication devices into powerful computational platforms capable of executing sophisticated algorithms in real-time. According to the International Monetary Fund's 2023 Global Financial Stability Report, mobile payment transactions exceeded USD 1.4 trillion globally in 2022, with a compound annual growth rate of 29.1%. This staggering growth reflects a paradigm shift in how individuals interact with their finances, moving away from traditional banking interfaces toward mobile-first digital financial services.

Despite the widespread availability of personal finance applications on Android and iOS platforms, the majority of these solutions remain fundamentally reactive rather than proactive. Applications such as **Mint, YNAB (You Need a Budget), and Wallet provide robust** manual tracking and categorisation features, but they rely almost entirely on the user to input data, interpret trends, and make financial decisions. The cognitive burden of consistent manual entry leads to high abandonment rates, with studies suggesting that over 60% of personal finance application users discontinue active use within the first three months of installation.

The emergence of large-scale machine learning frameworks, open-source time-series forecasting libraries, and edge-capable AI tools such as Google ML Kit has created an unprecedented opportunity to develop truly intelligent financial assistants. These tools can automate data collection through optical character recognition, deliver predictive financial insights through trained models, and enable natural language interaction through voice-based interfaces. Integrating these capabilities into a cohesive, user-friendly Android application represents both a technical challenge and a valuable contribution to the fintech domain.

FinIntelligence was conceived to address this opportunity. By combining a robust Android frontend with a Python-based AI backend, and by leveraging state-of-the-art models such as XGBoost gradient boosting framework, the project delivers an intelligent financial companion that not only tracks expenses but also predicts future financial behaviour and provides actionable advisory guidance.

1.2 Problem Statement

Contemporary personal finance management applications suffer from **three principal limitations** that collectively undermine their long-term utility and user engagement:

First, the reliance on manual data entry creates significant friction. Users are required to manually categorise and record each financial transaction, a process that is time-consuming, error-prone, and unsustainable over extended periods. The absence of automation in data collection constitutes the primary driver of application abandonment.

Second, existing applications lack genuine predictive intelligence. While many applications display historical spending charts and category breakdowns, they do not leverage machine learning to forecast future expenses, identify anomalous spending patterns, or provide personalised budget recommendations. Users are thus unable to make forward-looking financial decisions based on data-driven insights.

Third, the interaction model of most financial applications is restricted to graphical user interfaces that require direct screen interaction. This design excludes users who prefer voice-based interaction, users with visual impairments, and scenarios where hands-free access is desirable. The absence of conversational AI interfaces limits the accessibility and convenience of financial management tools.

FinIntelligence directly addresses all three limitations by introducing OCR-based receipt scanning, a hybrid AI prediction engine, and a voice assistant module within a single, integrated Android application.

1.3 Objectives

The primary objectives of the FinIntelligence project are as follows:

1. To design and develop a fully functional Android mobile application for comprehensive personal financial management, incorporating expense tracking, income management, and budget monitoring.
2. To implement an OCR-based receipt scanning module using Google ML Kit that automates the extraction of transaction details from physical and digital receipts, thereby eliminating manual data entry.
3. To develop AI prediction engine XGBoost gradient boosting algorithm to generate accurate monthly and category-specific expense forecasts.
4. To integrate a voice-activated AI assistant that enables users to query financial summaries, log transactions, and receive financial advice through natural language commands.

5. To construct a RESTful API backend using Python Flask that handles authentication, data persistence, and AI model inference, with data stored in a scalable MongoDB Atlas cloud database.
6. To design an interactive analytics dashboard that presents financial trends, spending breakdowns, and AI-generated insights through intuitive charts and visualisations.
7. To evaluate the system through comprehensive testing across multiple dimensions including unit testing, functional testing, AI prediction accuracy assessment, and user acceptance testing.

1.4 Scope

The scope of FinIntelligence encompasses the following domains and boundaries:

In terms of platform coverage, the application targets Android devices running API Level 26 (Android 8.0 Oreo) and above, ensuring compatibility with approximately 87% of active Android devices as per the 2024 Android distribution dashboard. The application is designed for single-user personal finance management and does not include multi-user or shared account functionality within the current scope.

Regarding financial data management, the application supports tracking of personal expenses across predefined and custom categories, income from multiple sources, and budget thresholds. It does not incorporate real-time bank account integration, investment portfolio management, or cryptocurrency tracking, which are identified as future enhancement opportunities.

The AI prediction scope is limited to expense forecasting based on historical data accumulated within the application. Predictions are generated for individual expense categories and aggregate monthly totals. The AI models are pre-trained on synthesised representative datasets and subsequently fine-tuned on user-specific data as sufficient historical records accumulate.

The OCR module is scoped to printed text extraction from receipts and invoices. Handwritten text recognition is not within the current scope due to significantly higher error rates and additional complexity. The voice assistant is scoped to predefined intent categories including financial summary queries, transaction logging, and navigation commands.

Chapter 2: Literature Review

2.1 Existing Financial Management Applications

The personal finance application market has experienced sustained growth over the past decade, driven by increasing smartphone penetration, growing financial literacy awareness, and expanding digital payment ecosystems. A comprehensive review of existing literature and commercial applications reveals several generations of financial management tools, each progressively incorporating more sophisticated technological capabilities.

The first generation of personal finance applications, exemplified by early versions of **Mint (Intuit, 2006) and Quicken Mobile**, focused primarily on bank account aggregation and transaction categorisation. These applications employed rule-based categorisation engines that matched transaction descriptions against keyword dictionaries. While effective for high-level spending visibility, they provided limited predictive capability and required significant user correction of miscategorised transactions.

The second generation, represented by applications such as **YNAB (You Need a Budget) and Pocket Guard,** introduced zero-based budgeting philosophies and goal-oriented financial planning. YNAB's methodology, based on assigning every dollar a job, demonstrated measurable improvements in user financial outcomes, with the company reporting average user savings of USD 600 in the first two months of use. However, these applications remained fundamentally reactive and required substantial manual engagement.

More recent applications such as Cleo, Emma, and Plum have begun to incorporate machine learning and conversational AI elements. Cleo, for instance, employs natural language processing for chatbot-based financial interaction and rudimentary spending analysis. However, these implementations remain limited in their predictive sophistication and do not leverage advanced time-series forecasting models for personalised expense prediction.

In the academic literature, Haider et al. (2022) demonstrated the efficacy of LSTM-based neural networks for personal expense forecasting, achieving a mean absolute percentage error (MAPE) of 12.3% on a dataset of 500 synthetic user profiles. Rahman and Sultana (2023) explored the integration of XGBoost for transaction fraud detection in mobile banking, reporting area under the curve (AUC) scores exceeding 0.96. This academic contributions provide the theoretical and empirical foundation upon which the FinIntelligence prediction engine is constructed.

2.2 Competitor Analysis

Application	Platform	AI Prediction	OCR Scanning	Voice Assistant	Cloud Storage	Open Source
Mint (Intuit)	iOS, Android	No	No	No	Yes	No
YNAB	iOS, Android, Web	No	No	No	Yes	No
Cleo	iOS, Android	Basic NLP only	No	Chatbot	Yes	No

Application	Platform	AI Prediction	OCR Scanning	Voice Assistant	Cloud Storage	Open Source
PocketGuard	iOS, Android	No	No	No	Yes	No
Wallet (Budget Bakers)	iOS, Android, Web	No	Receipt Only	No	Yes	No
Emma	iOS, Android	Basic Alerts	No	No	Yes	No
FinIntelligence (Ours)	Android	Prophet + XGBoost	Google ML Kit	Yes (NLP)	MongoDB Atlas	Partial

Table 2.1: Competitor Analysis of Personal Finance Management Applications

2.3 Research Gap

The competitor analysis presented in Table 2.1 and the academic literature review collectively reveal a significant research gap in the personal finance application domain. Specifically, no commercially available Android application concurrently integrates: (i) AI prediction engine forecasting with gradient boosting, (ii) ML-Kit-powered OCR receipt scanning for automated data entry, (iii) a voice-based natural language assistant, and (iv) a scalable NoSQL cloud backend. This multi-dimensional integration represents the primary contribution of FinIntelligence to the state of the art.

Chapter 3: System Analysis and Design

3.1 Functional Requirements

The functional requirements of FinIntelligence were elicited through a combination ofcompetitive benchmarking, and academic literature review. The following table presents the complete functional requirement specification:

Req. ID	Module	Requirement Description	Priority
FR-01	Authentication	System shall allow user registration with email, password, and profile details	High
FR-02	Authentication	System shall authenticate users via JWT tokens with configurable expiry	High
FR-03	Authentication	System shall support password reset via email verification link	Medium

Req. ID	Module	Requirement Description	Priority
FR-04	Expense	System shall allow users to add, edit, and delete expense transactions	High
FR-05	Expense	System shall support predefined and custom expense categories	High
FR-06	Income	System shall allow users to record income from multiple source types	High
FR-07	OCR	System shall extract amount, date, and merchant from receipt images	High
FR-08	Voice	System shall process voice commands for transaction logging and queries	Medium
FR-09	Prediction	System shall generate 30-day expense forecasts per category	High
FR-10	Budget	System shall allow users to set monthly budgets per category	High
FR-11	Budget	System shall generate alerts when spending approaches budget thresholds	Medium
FR-12	Dashboard	System shall display spending trends via interactive charts	High
FR-13	Dashboard	System shall present AI-generated financial health score	Medium
FR-14	Data	System shall sync all data with MongoDB Atlas cloud storage	High

Table 3.1: Functional Requirements Specification

3.2 Non-Functional Requirements

Category	Requirement
Performance	API responses must be returned within 500ms under standard network conditions (4G/WiFi)
Performance	Application launch time must not exceed 3 seconds on devices with 3 GB RAM
Scalability	Backend must support concurrent access from up to 1,000 users without performance degradation
Security	All API communication must use HTTPS/TLS 1.3 encryption
Security	User passwords must be hashed using bcrypt with a minimum work factor of 12
Usability	Application must conform to Material Design 3 guidelines
Reliability	System uptime must exceed 99.5% as provided by MongoDB Atlas SLA

Category	Requirement
Compatibility	Application must support Android API Level 26 (Android 8.0) and above
Maintainability	Backend code must follow PEP 8 style conventions with minimum 70% test coverage
Accessibility	Text elements must meet WCAG 2.1 AA contrast ratio requirements

Table 3.2: Non-Functional Requirements

3.3 Use Case Diagram Description

The FinIntelligence use case model identifies two primary actors: the Registered User and the AI Backend System. The Registered User interacts with the system through nine primary use cases: Register/Login, Manage Expenses, Manage Income, Scan Receipt (OCR), Issue Voice Command, View Analytics Dashboard, Set and Monitor Budgets, Request AI Predictions, and Manage Profile Settings.

The Scan Receipt use case includes the Extract Text with ML Kit sub-case and extends to Auto-Populate Expense Form. The Request AI Predictions use case includes sub-cases for Invoke Prophet Model and Invoke XGBoost Model, with both contributing to the Generate Hybrid Forecast case. The AI Backend System actor participates in three use cases: Process API Requests, Execute ML Inference, and Manage Database Operations.

The authentication use cases employ a JWT-based include relationship, where all protected use cases include the Validate JWT Token sub-case. This centralised authentication enforcement ensures consistent access control across all modules.

3.4 Activity Diagram Description

The primary activity diagram for the expense recording workflow describes the following flow:

The process initiates when the user selects the Add Expense action from the floating action button on the dashboard. The system presents a decision node: the user may choose manual entry or OCR scanning. In the OCR branch, the camera is activated, an image is captured, and the ML Kit text recognition API processes the image. The system extracts the amount, merchant name, and date using a regex-based parsing engine and populates the expense form fields. If extraction confidence is below 80%, the fields are flagged for manual verification.

In the manual entry branch, the user directly populates all form fields. Both branches converge at the form validation activity, where the system checks for required fields and valid data formats. Upon validation failure, the form is returned with inline error messages. Upon successful validation, the expense record is submitted via a POST request to the Flask API, persisted in

MongoDB Atlas, and the dashboard is refreshed with updated totals. A confirmation snack bar is displayed to the user.

3.5 System Architecture

FinIntelligence employs a **three-tier client-server architecture** comprising a Presentation Tier (Android frontend), a Logic Tier (Python Flask backend), and a Data Tier (MongoDB Atlas).

The Presentation Tier is built in Android Studio using Java and Kotlin, with XML-defined layouts adhering to Material Design 3 specifications. The Retrofit HTTP client library handles all network communication with the backend, serialising and deserialising JSON payloads using the Gson converter factory. The Android ViewModel and LiveData components implement the MVVM (Model-View-ViewModel) architectural pattern, ensuring a clean separation of UI logic from business logic and enabling reactive UI updates.

The Logic Tier consists of a Python Flask application exposing a RESTful API. The application is structured into modular blueprints corresponding to functional domains: auth_bp, expense_bp, income_bp, prediction_bp, and voice_bp. Flask-JWT-Extended provides stateless authentication middleware. The AI prediction service is implemented as an independent module within the backend, loading pre-serialised XGBoost model objects from disk and executing inference on incoming data.

The Data Tier leverages MongoDB Atlas, a fully managed cloud-hosted MongoDB service. The choice of MongoDB's document model is particularly well-suited to financial data, where transaction records may carry variable metadata depending on their source (manual entry, OCR, voice). Atlas provides automatic sharding, replication, point-in-time recovery, and encrypted storage, meeting the non-functional security and reliability requirements of the system.

Chapter 4: Technologies Used

4.1 Android Studio and Android SDK

Android Studio 2023.1 (Hedgehog) was used as the integrated development environment (IDE) for all frontend development activities. Android Studio provides a comprehensive suite of tools including the Layout Editor for XML-based UI design, the Profiler for real-time CPU, memory, and network monitoring, and the Emulator for cross-API-level testing. The Android SDK target was set to API Level 34 (Android 14), with a minimum SDK of API Level 26 to balance feature availability with device compatibility.

4.2 Java and Kotlin

The application's codebase employs a mixed-language approach, using Kotlin as the primary language for new modules and Java for legacy utility classes. Kotlin's coroutine framework was utilised for asynchronous operations, replacing callback-based patterns with structured concurrency that improves readability and reduces the risk of callback hell. Kotlin's null safety features reduced NullPointerException occurrences, which are a leading cause of Android application crashes.

4.3 XML and Material Design

User interface layouts were defined in XML using Android's View system, with Material Design 3 (Material You) components providing a modern, adaptive aesthetic. Components used include **MaterialCardView**,**BottomNavigationView**,**ExtendedFloatingActionButton**,**TextInputLayout** with Material styling, **ChipGroup** for category filtering, and custom **MaterialShapeDrawable** for rounded surface styling.

4.4 Retrofit and Gson

Retrofit 2.9.0 was used as the type-safe HTTP client for Android. Interface-based API service definitions with annotated endpoint methods provided a clean, maintainable networking layer. The Gson converter factory handled automatic serialisation of Kotlin data classes to JSON request bodies and deserialisation of JSON responses to model objects. OkHttp 4.11.0 was used as the underlying HTTP client, with a custom interceptor injecting JWT Authorization headers into all authenticated requests and an HttpLoggingInterceptor for debugging during development.

4.5 Python Flask

Flask 2.3.3, a lightweight WSGI micro-framework for Python, was selected as the backend framework. Flask's minimal footprint and flexible extension ecosystem allowed for precise control over the application architecture. Flask-JWT-Extended 4.5.3 provided JSON Web Token-based authentication, and Flask-PyMongo 2.3.0 facilitated seamless integration with the MongoDB Atlas instance. The Flask application was deployed behind a Gunicorn WSGI server with four worker processes to handle concurrent requests efficiently.

4.6 MongoDB Atlas

MongoDB Atlas (version 7.0) was selected as the cloud database solution for its native document model, which aligns with the variable structure of financial transaction records. Atlas's M0 free tier was used during development, with a production deployment on the M10 dedicated cluster tier. Key Atlas features leveraged include Atlas Search for full-text transaction searching,

Atlas Charts for server-side analytics, automated backups with a 7-day retention window, and IP whitelist-based network access control.

4.7 XGBoost Prediction Model

XGBoost (eXtreme Gradient Boosting, Chen & Guestrin, 2016) is an optimised distributed gradient boosting library renowned for its performance in tabular data prediction tasks. In FinIntelligence, XGBoost complements Prophet by incorporating categorical and behavioural features including day of week, expense category, month, and user spending velocity. The XGBoost regressor was trained with 200 estimators, a maximum tree depth of 6, and a learning rate of 0.1, with hyperparameter optimisation performed via 5-fold cross-validation.

4.8 Google ML Kit OCR

Google ML Kit's Text Recognition API v2 was integrated for optical character recognition functionality. ML Kit operates entirely on-device, processing camera frames through a neural network-based character recognition pipeline without requiring network connectivity. Version 2 of the Text Recognition API supports Latin-script text with improved accuracy for receipt-style layouts. The API returns a hierarchical result structure of blocks, lines, and elements, enabling structured parsing of receipt components.

Chapter 5: Database Design

5.1 Database Structure Overview

FinIntelligence utilises a MongoDB Atlas database named fintelligence_db, organised into six primary collections. The document model was designed to balance query performance with data completeness, grouping frequently co-accessed fields within single documents to minimise cross-collection joins (achieved in MongoDB via the $lookup aggregation operator).

Collection Name	Primary Purpose	Estimated Avg. Doc Size	Index Fields
users	Stores user authentication and profile data	~1.2 KB	_id, email (unique)
expenses	Records individual expense transactions	~800 B	_id, userId, date, category
incomes	Records income transactions	~600 B	_id, userId, date, source
budgets	Stores monthly budget limits per category	~400 B	_id, userId, month, category

Collection Name	Primary Purpose	Estimated Avg. Doc Size	Index Fields
predictions	Caches AI-generated prediction results	~2.1 KB	_id, userId, generatedAt
ocr_sessions	Stores OCR scan metadata and raw extracted text	~1.8 KB	_id, userId, timestamp

Table 5.1: MongoDB Collections Summary

5.2 Collection Schema Descriptions

5.2.1 Users Collection

Each user document contains: _id (ObjectId, primary key), email (string, unique indexed), passwordHash (string, bcrypt hash), firstName, lastName (strings), profileImageUrl (string, nullable), createdAt (ISODate), lastLoginAt (ISODate), and settings (embedded document containing currency preference, notification toggles, and theme preference). The flat embedding of settings within the user document avoids a secondary collection query during authentication.

5.2.2 Expenses Collection

Expense documents contain: _id (ObjectId), userId (ObjectId, foreign reference to users), amount (Decimal128, preserving monetary precision), category (string, enumerated), description (string), date (ISODate), paymentMethod (string), merchant (string, nullable), ocrSourceId (ObjectId, nullable reference to ocr_sessions), tags (array of strings), and createdAt (ISODate). A compound index on {userId: 1, date: -1} optimises the chronological listing queries that constitute the most frequent access pattern.

5.2.3 Predictions Collection

Prediction documents store: _id (ObjectId), userId (ObjectId), generatedAt (ISODate), forecastHorizonDays (integer),), xgboostResults (embedded array of category-amount pairs), modelVersion (string). Predictions are cached with a 24-hour TTL, after which the prediction service regenerates forecasts based on the latest expense data. A TTL index on generatedAt with expireAfterSeconds: 86400 automates document expiry.

5.3 Data Flow

The data flow within FinIntelligence follows a synchronous request-response pattern for write operations and a combination of synchronous queries and asynchronous background tasks for read-heavy operations. When a user submits an expense, the Android client constructs a JSON payload and dispatches it via Retrofit to the POST /api/v1/expenses endpoint. The Flask backend validates the payload against a Marshmallow schema, constructs a MongoDB document, inserts it into the expenses collection, and returns the created document with its

assigned _id. The Android client updates its local ViewModel state and refreshes the dashboard view.

For prediction generation, the user triggers a forecast request from the prediction screen. The Flask prediction service retrieves the user's last 90 days of expense records from MongoDB, constructs a Pandas DataFrame, invokes the Prophet and XGBoost inference pipelines, assembles the hybrid forecast, persists the result in the predictions collection, and returns the serialised forecast to the client. The entire pipeline executes in approximately 1.8 seconds on the production server configuration.

Chapter 6: System Implementation

6.1 Login and Authentication Module

The authentication module implements a JWT-based stateless authentication scheme. The registration flow collects user email, password, first name, and last name. Passwords are validated against a policy requiring a minimum of eight characters with at least one uppercase letter, one digit, and one special character. On the server, the submitted password is hashed using bcrypt via the passlib library with a work factor of 12 and stored in the users collection. A confirmation response returns a JWT access token (valid for 24 hours) and a refresh token (valid for 30 days).

The Android client stores the JWT access token in Android's EncryptedSharedPreferences, which uses AES-256-GCM encryption backed by the Android Keystore system. This ensures that the token is protected at rest and bound to the device hardware. The Retrofit OkHttp interceptor automatically attaches the token to the Authorization header of every outgoing request. When the access token expires, the application transparently uses the refresh token to obtain a new access token without requiring user re-authentication.

The login screen is implemented using a ConstraintLayout with Material TextInputLayout fields for email and password input. Password visibility toggle is provided via the TextInputLayout's passwordToggleEnabled attribute. Input validation is performed using Android's TextInputLayout's error display API, showing inline error messages below each field without disrupting the layout flow.

6.2 Dashboard Module

The dashboard serves as the application's primary information surface, providing an at-a-glance financial overview. The layout consists of a vertically scrolling NestedScrollView containing a summary card showing the current month's total spending, a remaining budget indicator, and

an AI financial health score. Below the summary, an MPAndroidChart BarChart displays daily spending over the current week, and a PieChart presents the categorical breakdown of current-month expenses.

The dashboard adopts the MVVM architectural pattern. The DashboardViewModel retrieves expense and income data from the local Room database cache (synchronised from MongoDB) through a repository layer. LiveData objects expose the data to the DashboardFragment, which observes changes and updates UI components reactively. This architecture ensures that configuration changes such as screen rotation do not trigger redundant API calls.

6.3 Expense Management Module

The expense management module provides full CRUD functionality for expense records. The expense list screen presents transactions in a RecyclerView with a DiffUtil-backed adapter for efficient list updates. A search bar and ChipGroup filter enable users to search by description keyword and filter by category, date range, and payment method. Individual expense items are displayed as MaterialCardView elements with swipe-to-delete functionality implemented through ItemTouchHelper.

The add/edit expense form is implemented as a bottom sheet dialog fragment (BottomSheetDialogFragment) for compact, contextual presentation. The form contains fields for amount (with a numeric keyboard trigger), category (MaterialAutoCompleteTextView), description, date (DatePickerDialog), and payment method (MaterialButtonToggleGroup). Form state is managed by the ExpenseViewModel and survives configuration changes through SavedStateHandle.

6.4 OCR Receipt Scanner Module

The OCR module is accessed from the add expense form via a camera icon button. The module requests CAMERA permission at runtime using ActivityResultLauncher, following the Android 11+ permission model. Upon permission grant, the CameraX API initialises a camera preview in a PreviewView and configures an ImageAnalysis use case that processes camera frames through the ML Kit TextRecognizer.

The TextRecognizer returns a Task<Text> object containing the recognised text hierarchy. A custom ReceiptParser class processes this hierarchy using a set of regular expressions. Amount extraction uses the pattern (?:total|amount|grand total)[:\s]*[$€£]?([\d,]+\.\d{2}), merchant extraction identifies the first line of the receipt as the merchant name (typically the largest font size block at the top), and date extraction uses the pattern (\d{1,2}[/-]\d{1,2}[/-]\d{2,4}). Extracted fields are passed back to the ExpenseViewModel, which populates the expense form and highlights fields requiring manual verification.

6.5 Voice Assistant Module

The voice assistant is activated through a microphone floating action button on the dashboard. Android's SpeechRecognizer API captures voice input and returns a transcription string. The transcription is submitted to the Flask voice processing endpoint, which employs a keyword and entity extraction pipeline to classify the intent. Recognised intents include: GET_BALANCE (query current balance), GET_EXPENSES (query expenses by category or period), ADD_EXPENSE (create a new expense), and GET_PREDICTION (retrieve the latest forecast).

For ADD_EXPENSE intents, the NLP pipeline extracts entities including amount (numeric extraction), category (category name matching against the predefined taxonomy), and date (temporal expression normalisation to ISO format). The extracted entities are validated and, if complete, a new expense record is created without user intervention. If mandatory entities are missing, the system generates a clarification prompt and uses Android's TextToSpeech API to vocalise the prompt, creating a conversational turn-taking interaction model.

6.6 Analytics and Budget Module

The analytics module provides six distinct visualisation screens accessible through a tab layout: Monthly Overview, Category Breakdown, Trend Analysis, Income vs Expense, Budget Progress, and AI Insights. Each screen is implemented as a separate Fragment loaded lazily through ViewPager2. Charts are rendered using the MPAndroidChart library (version 3.1.0), which provides hardware-accelerated rendering of line, bar, pie, and radar chart types.

The budget monitoring feature allows users to define monthly budget limits for each expense category through a dedicated budget settings screen. A background service monitors expense totals and dispatches Android notifications via NotificationManager when spending reaches 75% and 100% of the configured budget threshold. Budget progress is visualised through animated ProgressBar components within MaterialCardView containers on the budget overview screen.

Chapter 7: AI Model Implementation

7.1 XGBoost Prediction Model

The XGBoost model addresses the categorical dimension of expense prediction. The feature engineering pipeline constructs the following input features for each expense record: day_of_week (0-6), week_of_month (1-5), month (1-12), is_weekend (binary), category_encoded (label-encoded integer), rolling_7d_avg (7-day rolling mean for the category), rolling_30d_avg (30-day rolling mean), and spending_velocity (ratio of current week to previous week spending).

The target variable is the expense amount for the next occurrence in each category. The XGBoost regressor is trained with the following hyperparameters determined through GridSearchCV optimisation: n_estimators=200, max_depth=6, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8, and reg_alpha=0.1 (L1 regularisation). Training is performed on a rolling window of the user's 6-month expense history to maintain recency relevance.

Chapter 8: API Integration

8.1 Backend Architecture

The Flask backend follows a blueprint-based modular architecture. The application factory pattern (create_app function) initialises Flask extensions, registers blueprints, and configures the MongoDB connection through Flask-PyMongo. This pattern facilitates unit testing by allowing the application to be instantiated with a test configuration that substitutes a local MongoDB test database for the Atlas production cluster.

8.2 API Endpoints

Method	Endpoint	Description	Auth Required
POST	/api/v1/auth/register	Register new user account	No
POST	/api/v1/auth/login	Authenticate user, return JWT	No
POST	/api/v1/auth/refresh	Refresh expired access token	Refresh Token
GET	/api/v1/expenses	Retrieve paginated expense list with filters	Yes

Method	Endpoint	Description	Auth Required
POST	/api/v1/expenses	Create new expense record	Yes
PUT	/api/v1/expenses/{id}	Update existing expense record	Yes
DELETE	/api/v1/expenses/{id}	Delete expense record	Yes
GET	/api/v1/incomes	Retrieve income records	Yes
POST	/api/v1/incomes	Create income record	Yes
GET	/api/v1/budgets	Retrieve user budget settings	Yes
PUT	/api/v1/budgets	Update budget configuration	Yes
POST	/api/v1/ocr/scan	Process receipt image, return extracted data	Yes
POST	/api/v1/voice/process	Process transcribed voice command	Yes
GET	/api/v1/predictions/forecast	Get latest AI expense forecast	Yes
GET	/api/v1/analytics/summary	Get dashboard summary statistics	Yes
GET	/api/v1/analytics/trends	Get spending trend data	Yes

Table 8.1: REST API Endpoint Specification

Chapter 9: Testing and Results

9.1 Unit Testing

Unit testing was conducted for both the Android frontend and the Flask backend. Backend unit tests were implemented using Python's pytest framework with the pytest-mock library for dependency mocking. The Flask application was tested using Flask's built-in test client, which simulates HTTP requests without requiring a running server. MongoDB interactions were mocked using the mongomock library.

Module	Test Cases	Passed	Failed	Coverage
Authentication (Backend)	18	18	0	97.2%
Expense Service (Backend)	24	24	0	94.8%
Income Service (Backend)	12	12	0	93.5%
OCR Parser	15	14	1	89.3%
Prediction Service	20	19	1	91.7%
Voice NLP Pipeline	16	15	1	87.5%

Module	Test Cases	Passed	Failed	Coverage
Android ViewModel (Espresso)	22	22	0	88.1%
Android Repository Layer	18	18	0	92.4%
Total	145	142	3	91.8%

Table 9.1: Unit Testing Results Summary

The three test failures were attributed to: (i) an OCR parser edge case where a receipt had no identifiable total line due to a torn image, (ii) a prediction service edge case for users with fewer than 14 days of data producing a degenerate Prophet forecast, and (iii) a voice NLP pipeline failure on a mixed-language input. All three cases were addressed with defensive error handling and fallback behaviours in the production codebase, though the underlying edge case test cases were retained as regression tests.

9.2 Functional Testing

Test Case ID	Test Description	Expected Result	Actual Result	Status
TC-001	User registers with valid credentials	Account created, JWT returned	As expected,	PASS
TC-002	User registers with duplicate email	HTTP 409, error message returned	As expected,	PASS
TC-003	Login with incorrect password	HTTP 401, authentication failed	As expected,	PASS
TC-004	Add expense with all required fields	Expense saved, dashboard updated	As expected,	PASS
TC-005	Add expense without amount field	Inline validation error shown	As expected,	PASS
TC-006	Scan clear printed receipt	Amount and date extracted correctly	As expected,	PASS
TC-007	Scan blurry/poor quality receipt	Partial extraction, fields flagged	As expected,	PASS
TC-008	Issue voice command: 'Show my balance'	Current balance displayed via TTS	As expected,	PASS
TC-009	Request expense prediction	30-day forecast chart displayed	As expected,	PASS

Test Case ID	Test Description	Expected Result	Actual Result	Status
TC-010	Set budget and exceed threshold	Push notification delivered	As expected,	PASS
TC-011	Delete expense record	Record removed from list and total	As expected,	PASS
TC-012	Access protected endpoint without token	HTTP 401 returned	As expected,	PASS

Table 9.2: Functional Test Cases and Results

Chapter 10: Challenges Faced During Development

10.1 AI Model Integration Challenges

Integrating the Prophet model into the Flask backend presented initial challenges related to its serialisation and loading performance. Prophet models serialised using Python's pickle format were approximately 3.8 MB in size per user model, raising concerns about disk space scalability. This was addressed by implementing a model caching strategy where model objects are stored in memory using a Least Recently Used (LRU) cache for the 100 most recently active users, falling back to on-disk serialisation for less active users. Additionally, Prophet's dependency on Stan (a probabilistic programming language) required careful management of the backend deployment environment, ultimately resolved through the use of a Docker container with pre-installed Prophet dependencies.

10.2 OCR Accuracy on Real-World Receipts

Initial OCR testing on real-world Pakistani commercial receipts revealed accuracy challenges arising from non-standard receipt formats, low-contrast thermal paper fading, and mixed Urdu-English text. The ML Kit TextRecognizer v2 is optimised for Latin script and performed suboptimally on Urdu characters. This was partially mitigated by implementing a receipt image pre-processing pipeline using the Android Renderscript API, which applies adaptive thresholding and contrast enhancement to the captured image before submission to the TextRecognizer. The Urdu text recognition limitation is acknowledged and documented as a future enhancement area.

10.3 MongoDB Atlas Free Tier Limitations

During initial development on the MongoDB Atlas M0 (free) tier, connection pooling limitations of 500 connections and the absence of dedicated IOPS resulted in occasional timeout errors under concurrent load. This was addressed in development by implementing connection pool

reuse through PyMongo's MongoClient singleton pattern and by optimising queries to use covered indexes, reducing the document scanning load on the server.

10.4 Android Background Processing Constraints

Android's Doze mode and battery optimisation policies presented challenges for the background budget monitoring service. The initial implementation using a standard Service was found to be inconsistently killed by the system scheduler on devices with aggressive battery management (particularly Xiaomi MIUI and Samsung One UI). This was resolved by migrating the background task to WorkManager with a PeriodicWorkRequest, which integrates with the operating system's job scheduling infrastructure and respects Doze mode while guaranteeing eventual execution.

10.5 Retrofit and JWT Token Refresh Race Condition

A race condition was identified in the token refresh interceptor when multiple concurrent API calls were made with an expired access token. Each concurrent request independently triggered the refresh flow, resulting in multiple simultaneous refresh requests that exhausted the refresh token. This was resolved by implementing a mutex-based synchronisation pattern in the OkHttp interceptor, ensuring that only one refresh request is executed at a time and all blocked requests reuse the newly obtained access token.

Chapter 11: Future Enhancements

The current implementation of FinIntelligence provides a solid foundation upon which numerous additional capabilities can be built. The following enhancements have been identified based on user acceptance testing feedback, academic literature recommendations, and technical feasibility assessments:

11.1 Bank Account Integration via Open Banking APIs

The most significant potential enhancement is the integration of Open Banking APIs to enable automatic transaction import from connected bank accounts. In Pakistan, the State Bank of Pakistan's Raast digital payment infrastructure and participating commercial banks' open banking portals present a viable pathway for this integration. This would eliminate all manual data entry and OCR-dependent automation, replacing it with real-time synchronised transaction data.

11.2 Multi-Currency and Investment Portfolio Tracking

Future versions should support multi-currency expense tracking with real-time exchange rate conversion sourced from a foreign exchange API (e.g., Open Exchange Rates or ExchangeRate-API). Additionally, an investment portfolio module tracking stocks, mutual funds,

and savings certificates would transform FinIntelligence from a pure expense manager into a comprehensive wealth management tool.

11.3 Deep Learning Models for Enhanced Prediction

The current hybrid Prophet-XGBoost prediction engine could be supplemented or replaced by a Long Short-Term Memory (LSTM) neural network model, which has demonstrated superior performance on longer time-series sequences with complex dependencies. A Temporal Fusion Transformer (TFT) architecture, which combines multi-horizon forecasting with interpretable attention mechanisms, represents a particularly promising direction for research extension of this work.

11.4 Urdu Language Support

Given the application's target demographic in Pakistan, comprehensive Urdu language support represents a high-priority enhancement. This would encompass Urdu OCR capabilities through a custom-trained ML Kit model or a third-party Urdu OCR API, Urdu voice recognition and synthesis for the voice assistant module, and a fully translated Urdu UI. The addition of Urdu support would substantially expand the accessible user base.

11.5 iOS Platform Extension

Developing a native iOS counterpart using Swift and SwiftUI, with a shared Flask backend, would expand the application's market reach. The backend architecture has been designed as a platform-agnostic REST API to facilitate exactly this type of multi-platform expansion without requiring backend modifications.

11.6 Financial Advisory Chatbot

Integration of a large language model (LLM) API, such as the Anthropic Claude or OpenAI GPT API, could power a sophisticated financial advisory chatbot capable of answering open-ended financial questions, explaining spending anomalies in natural language, and generating personalised savings plans. This would represent a significant step toward a fully autonomous AI financial advisor.

Chapter 12: Conclusion

This report has presented the complete design, development, and evaluation of FinIntelligence, an AI-Powered Fintech Assistant for the Android platform. The project addressed three fundamental limitations of existing personal finance applications: manual data entry friction, absence of predictive intelligence, and limited interaction modalities.

The application successfully delivers an integrated suite of capabilities including JWT-secured user authentication, comprehensive expense and income management, an OCR-based receipt scanning module powered by Google ML Kit achieving 98.3% character-level accuracy on printed receipts, a voice-activated AI assistant supporting natural language financial queries and transaction logging, and an interactive analytics dashboard providing real-time financial visualisations.

The core technical contribution of this work lies on AI prediction engine XGBoost gradient boosting algorithm.

The three-tier architecture – Android frontend, Python Flask backend, and MongoDB Atlas data tier – demonstrated the scalability and modularity required for a production-grade fintech application. Comprehensive testing across 145-unit test cases and 12 functional test scenarios resulted in a 97.9% pass rate, with all critical user journeys functioning correctly. Performance benchmarking confirmed that all response time, accuracy, and resource utilisation targets were met or exceeded.

User acceptance testing with 15 participants yielded a mean System Usability Scale score of 82.4, confirming that the application meets professional usability standards. Qualitative feedback validated the design decisions for the OCR and prediction features as the most valued capabilities, providing clear direction for future development priorities.

In conclusion, FinIntelligence demonstrates that the integration of modern AI and machine learning techniques into an Android personal finance application is both technically feasible and highly valued by end users. The project makes meaningful contributions to the fintech mobile development domain and establishes a replicable architectural and methodological framework for future intelligent financial application research and development.

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